

Department of Mathematics, TIET, Patiala
Differential Equations and Linear Algebra (UMA023)
Tutorial Sheet - 02
Second Order Linear Homogeneous Differential Equations

1. If $y_1(x)$ is solution of the given differential equation, then find the other solution $y_2(x)$ and hence the general solution.

(a) $y'' - 4y' + 4y = 0$; $y_1 = e^{2x}$

(b) $x^2y'' - 3xy' + 4y = 0$; $y_1 = x^2 \ln x$

(c) $x^2y'' - 7xy' + 16y = 0$; $y_1 = x^4$

(d) $x^2y'' - xy' + 2y = 0$; $y_1 = x \sin(\ln x)$

(e) $(1 - x^2)y'' + 2xy' = 0$; $y_1 = 1$

(f) $(1 - 2x - x^2)y'' + 2(1 + x)y' - 2y = 0$; $y_1 = x + 1$

2. Determine the general solution of the following differential equations.

(a) $y'' - y' - 6y = 0$

(b) $y'' + 8y' + 16y = 0$

(c) $12y'' - 5y' - 2y = 0$

(d) $y'' + 9y = 0$

(e) $y'' - 4y' + 5y = 0$

(f) $4y'' + y' = 0$

3. Solve the following differential equations (Cauchy - Euler equation)

(a) $x^2y'' - 2y = 0$

(b) $xy'' + y' = 0$

(c) $x^2y'' + xy' + 4y = 0$

(d) $x^2y'' - 3xy' - 2y = 0$

(e) $x^2y'' - 3xy' + 4y = 0$

(f) $3x^2y'' + 6xy' + y = 0$

Answers

1. (a) $y_2 = xe^{2x}$

(b) $y_2 = -x^2$

(c) $y_2 = x^4 \ln x$

(d) $y_2 = -x \cos(\ln x)$

(e) $y_2 = x - \frac{x^3}{3}$

(f) $y_2 = x^2 + x + 2$

2. (a) $y = c_1 e^{3x} + c_2 e^{-2x}$

(b) $y = (c_1 + c_2 x)e^{-4x}$

(c) $y = c_1 e^{2x/3} + c_2 e^{-x/4}$

(d) $y = c_1 \cos 3x + c_2 \sin 3x$

(e) $y = e^{2x}(c_1 \cos x + c_2 \sin x)$

(f) $y = c_1 + c_2 e^{-x/4}$

3. (a) $y = c_1 x^{-1} + c_2 x^2$

(b) $y = c_1 + c_2 \ln x$

(c) $y = c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)$

(d) $y = c_1 x^{2-\sqrt{6}} + c_2 x^{2+\sqrt{6}}$

(e) $y = (c_1 + c_2 \ln x)x^2$

(f) $y = x^{-\frac{1}{2}} \left(c_1 \cos \left(\frac{\sqrt{3} \ln x}{6} \right) + c_2 \sin \left(\frac{\sqrt{3} \ln x}{6} \right) \right)$